GraviFi

Every market, in one orbit.

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Abstract

GraviFi turns a mix of tokenized stocks, crypto and memecoins into a single token. Each **basket** is a vault on Solana that holds its tokens and issues one share token for them. A share is always worth its exact slice of what the vault holds, and can always be swapped back for that slice. Anyone can buy a basket with USDC or SOL, or build one from 133 listed assets. Every basket mint and redeem pays a 0.40% protocol fee; half of it buys **GRAVI** on the open market and burns it. Creating a basket burns GRAVI too.

GraviFi has been live on Solana mainnet since 14 October 2026.

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1. The problem

Tokenized stocks, pre-IPO tokens, crypto and memecoins now trade side by side on Solana, but holding a spread of them is still manual: a dozen swaps to buy in, a dozen to rebalance, a dozen to get out, and a wallet full of positions to track. Traditional index funds solve this off-chain, behind a broker, a custodian and a fund manager, and they can't hold a memecoin or a token that isn't listed on an exchange.

GraviFi does it on-chain: one token for the whole spread, held in your own wallet, with the rules written into the program rather than into a prospectus.

2. Baskets

A basket is three things:

- **A vault.** The GraviFi program holds the basket's tokens in accounts only it controls. No person, not the creator and not the team, can move them outside the rules below.
- **A share token.** One token for the whole basket. It lives in the holder's wallet like any other Solana token and can be sent or traded freely.
- **A recipe.** The basket records how many units of each token back one share. A share's value (its NAV) is those units priced live.

2.1 MINTING AND REDEEMING

Minting deposits the exact underlying tokens for a number of shares and receives those shares, less the protocol fee. **Redeeming** hands shares back and receives each token's exact slice, also less the fee. No price is involved in either: the vault swaps units for units, so a share can never be issued for less than it is worth, or paid out for more.

Both are priced from what the vault holds at that moment, not from a figure stored earlier, so tokens that reach a vault any other way (an airdrop, a donation, a token a redeemer left behind) belong to the holders already there: nobody can mint into them at the old price.

The vault always rounds in its own favour: deposits round up, payouts round down. Every token a share promises is in the vault.

2.2 REDEMPTION IS ALWAYS OPEN

Redeeming is never gated: not by a basket's access mode, not by its creator, not by a delisting, not by the team. If the issuer of one token pauses or freezes it, a holder can leave that one token behind and take everything else; what they leave stays with the remaining holders. **A holder who cannot redeem holds a promise, not a claim**, so GraviFi is built so that this never happens.

2.3 BUYING AND SELLING WITH USDC OR SOL

Nobody needs to hold the underlying tokens first. The app turns a payment into a mint:

1. **Swaps.** The payment is split by each token's weight and swapped through Jupiter into the buyer's own wallet. Each swap carries a price limit.

2. **Mint.** The vault takes what the swaps delivered and mints the shares it covers. If a swap misses its limit, nothing is minted, and the app offers to swap back what landed.

Selling runs the other way: redeem, then swap each token back to USDC or SOL. Or redeem and keep the tokens. Every step is approved in the buyer's wallet, and the full cost is shown before signing.

3. Creating a basket

Anyone can create a basket in the app:

CHOICE	OPTIONS
Assets	2 to 14, from the 133 listed
Weights	Any split adding up to 100%
Rebalancing	On a schedule, when it drifts past a tolerance, or both (at least a day apart, at least 0.5% drift). Rebalancing starts once the basket holds $50 (section 4)
Creator's cut	0% to 0.60% of each buy
Access	Public; allowlist (only wallets the creator approves can mint); or private (only the creator can mint). Anyone holding shares can always redeem
Control	Creator-managed, or immutable: locked forever, nobody can change it again
Starting price	$1 to $10,000 a share

Creating the basket burns the GRAVI the protocol asks for (section 6.3); the creator then seeds it with at least $20, and its shares go live on the app's explore page. A creator-managed basket can later change its weights, add or remove assets, change its rebalancing policy and fee within the caps, and split its shares.

Splits. A creator can split or reverse-split a basket's shares (1:100 to 100:1, at most once every 30 days). Every holder's balance and the price per share change together in one transaction; nobody's slice changes.

The creator's cut is a share of each buy by someone else, paid in shares. Selling and holding don't pay it, so nobody already in can be charged more by a later rise. A creator can lower it at any time. Raising it takes a day's notice once the basket has shares, shown on its page, so no buyer pays more than they were quoted. An immutable basket's is fixed at its first deposit.

The yearly fee is how creators earned before the cut: up to 0.20% a year, streamed as new shares while held. Baskets that have one keep it until their creator chooses a cut, which stops it. A basket has one or the other, never both, and new baskets are only offered the cut.

4. Rebalancing

Prices move, so a basket's actual weights drift from its targets. When a basket's schedule comes round, or it drifts past its tolerance, a keeper trades it back:

- **In slices** of at most 5% of the basket's value, so no single trade moves a thin market. An asset on its way out (set to 0% by its creator, or stopped out, section 4.1) sells in clips of up to 50%, each still stopping at the bought asset's target.
- **At unpredictable times**, so nobody can trade ahead of a known schedule.
- **Against a price floor.** Each slice must fetch at least the oracle price, less a small tolerance (1% for stocks and crypto; up to 3%, the program's ceiling, when a memecoin is involved, whose pool fees alone can pass 1.5%), or the whole slice reverts. The keeper is lent the tokens to sell, and the program checks what comes back to the vault in the same transaction.

The program checks every slice, and the share supply can't change in the middle of one. The floor is only as good as the price it's checked against. A price service posts those prices with its own key, the program's price authority: it reads each price from Jupiter itself, so the app can ask it to post a price but never choose one. The program bounds what it may post. Where an asset has a Pyth feed (about 20 of them today: SOL, ETH, BTC, JUP, BONK, WIF and others), a posted price must be within 3-5% of a fresh, fully verified Pyth price. Any other asset's price may move at most 20% from the last accepted one per hour, so a false price can't appear at once. Within those bounds a rebalance can't take value out of a basket as long as the prices posted are honest. Section 8 says who holds that key.

Only baskets holding at least $50. The keeper doesn't rebalance a basket worth less than $50: nothing is traded, not for drift, not on schedule, not even an asset its creator is adding or removing. Its holdings drift with the market until it grows past $50; minting and redeeming work

as usual. Below that a slice is a few cents, and hundreds of small baskets would take turns from the real ones. Each basket's page says when it's below the line.

The keeper is paid back, not paid. Running a basket costs the keeper network fees: each slice, the prices posted for it, and once, the basket's rebalance record. Each slice pays it back what it spent on that basket so far, in the token the slice sells, and at most $0.05 a slice, a limit in the program's code. What isn't covered yet waits for the next slice. The keeper sells those tokens for SOL. Holders pay that cost as a sliver of the sold token, and nobody earns anything from it. Sweeps and the buyback pay back the same way, the buyback from the tokens it sells for GRAVI.

Taking turns. The keeper runs a few slices an hour, shared by every basket. The baskets that want a slice go in order of how long they've waited since their last one: a basket goes to the back of the queue each time it's served, or when its slice fails. So a volatile basket that's due every hour can't take the turns of the others, and every basket gets one in its turn.

Tokens a basket is sent. Some tokens pay their holders in another token (a stonkfun token can pay ZEC), and anyone can send a basket an airdrop. Those tokens aren't part of the recipe, so the keeper sweeps them: it swaps them into the basket's most underweight token against the same oracle floor, and their value goes to the holders.

A token the market stops pricing. An asset at 0% whose price hasn't been posted for a day counts as worth nothing in the books, so it can't block the rest of the basket. Nothing can be sold or bought at $0, so it stays in the vault and redeeming still pays it out.

4.1 STOP LOSSES

A creator can set a stop loss on any of a basket's assets: a percentage, 5% to 90%, below a reference price. A fixed stop's reference is the asset's price when the stop was set ("down 30% from now"); a trailing stop's reference rises with every higher price, so it sits below the asset's highest price since. The stops are stored on chain, where anyone can read them.

If the asset's posted price is below its stop, the stop arms; if it's still below 5 minutes later, it fires. The program checks both prices (each fresh, within the bounds of section 4) and the time between them, and anyone may ask it to: the keeper does every minute. A price back above the stop before then disarms it. Firing sets the asset's target to 0% and scales every other target up in proportion, so the basket is at 100% again without it, and the stop is gone. The keeper then sells the asset out in clips of up to half the basket, a minute apart, each into the asset furthest

under its new target and each held to the oracle floor. This happens whatever the basket's size: the $50 line doesn't apply to a stop.

What a stop can't do: it doesn't buy back if the price recovers; a creator can add the asset again. A stop fires on the posted price, so on an asset without a Pyth feed, whose price may fall at most 20% an hour, a stop after a crash fires and sells in steps, over hours; the app labels these "slow stops". A thin token can be pushed through its stop for 5 minutes, and a stop sells near the bottom of a dip. An immutable basket's stops are part of its recipe: they're set before its first deposit and never change, and when one fires the basket follows that fixed rule.

5. Fees

FEE	RATE	WHO PAYS, WHEN	WHERE IT GOES
Protocol fee	0.40%	On every mint and redeem, including a creator's first deposit	Half to the GRAVI burn, half to the treasury
Creator's cut	0% to 0.60%	On each buy by someone else, on top of the protocol fee. Selling and holding don't pay it	The basket's creator
Yearly fee	0% to 0.20% a year	Holders, as share dilution while held, on baskets that haven't chosen a cut	The basket's creator
Creation burn	10,000 GRAVI per asset	The creator, for each asset a basket is created with or gets added	Burned
App fee	0.05%	On buys and sells through the app's USDC/SOL route	The treasury
Network	fractions of a cent	Every transaction	Solana validators

Invited wallets. A wallet that joins with someone's invite doesn't pay the GRAVI-burn half of the protocol fee for its first 30 days: it pays 0.20%, all of it to the treasury. The app fee is unchanged. The program reads the wallet's invite record, good for at most 31 days, which the wallet writes with its first trade and the keeper co-signs. The wallet puts down the record's rent, under 0.001 SOL, and gets it back with its first trade after the 30 days. For a year, the inviter receives half of the treasury's part of the protocol fee on that wallet's basket trades (0.10% of each trade),

worked out in SOL at each trade's price and paid daily in SOL by the app. It is a share of fees from someone else's trading, not a return on anything held, and it never pays out more than those trades paid in.

The protocol fee rounds up, so no flow is too small to pay it, and it is kept in shares, so the vault stays fully backed. A creator's cut rounds up too and waits in the treasury's shares until it's claimed, by anyone, to the creator's own account; the treasury can't withdraw it. The limits are in the program's code: the protocol fee can never exceed 0.50%, a creator's cut can never exceed 0.60% of a buy, a yearly fee can never exceed 0.20%, the creation burn can never exceed 1,000,000 GRAVI per asset, and at least half of every protocol fee goes to the GRAVI burn, except an invited wallet's half fee in its first month (above). Changing any of them would take new code, in public.

Opening a token account the first time a wallet holds an asset costs about 0.002 SOL of Solana rent, refunded when the account is closed. Trading shares wallet to wallet, or in any pool someone else runs, pays the protocol nothing.

6. GRAVI

GRAVI is GraviFi's token. It carries no vote and no claim on baskets; nothing to lock, nothing to stake. Its one job is to absorb the protocol's revenue through burns: the more GraviFi is used, the less GRAVI there is.

Address	`TRGNtERnbRtVkaMJ1MqU4DzSEkHrtd2qFesh6Rgravi`
Supply	1,000,000,000 at launch, fixed: nobody can mint more
Launch	14 October 2026, on pump.fun, paired with SOL
Team	1.92% of supply (19,221,405 GRAVI), locked on Streamflow until 15 April 2027: one release, no cancel, no transfer

6.1 BUYBACK AND BURN

Half of every protocol fee is set aside in the **burn address** (`FkMEzsnij3aN9njxcjmGxi1DK43WChBtRWorvXgptUJr`). It has no private key: it is controlled by the GraviFi program, and the program can only spend it one way.

1. The burn address redeems its basket shares for the underlying tokens, exactly as any holder would.
2. The keeper sells those tokens for GRAVI on the open market. The program checks the GRAVI received against the oracle price before accepting it.
3. All of the GRAVI received is burned in the same transaction.

Buybacks run in slices once there is enough to be worth the network fees, so small amounts wait until they add up. Every burn is public on Solana, and the running total is shown live at app.gravifi.xyz/pulse.

6.2 GRAVI'S OWN TRADING FEES

GRAVI's pump.fun creator fees go the same way: claimed into the team's GRAVI wallet, half buys GRAVI and burns it, half goes to the treasury.

Burn #1 (15 October 2026): 3.80 SOL of creator fees claimed; 1.90 SOL bought and burned **10,611,685 GRAVI**, 1.06% of supply. Supply after: 989,388,314.

6.3 CREATION BURN

Every asset in a basket burns GRAVI from the creator's wallet: creating a basket burns it for each asset the basket starts with, in the instruction that creates it (a basket created and never seeded has still paid it), and adding an asset later burns it for that asset. It is a real burn, so supply falls; nobody receives it. The amount is **10,000 GRAVI per asset** today (100,000 until 25 October 2026), so a five-asset basket burns 50,000. The protocol authority sets it (section 8), and 0 turns it off; the program refuses anything above **1,000,000 GRAVI per asset**, and only a program upgrade could raise that ceiling. A creator doesn't need to hold GRAVI first: the app buys whatever is missing with SOL as part of creating or adding, and shows the cost before signing.

One wallet can be exempt: the protocol authority can name a single team wallet whose baskets burn nothing, so the team can launch its own promotional baskets without spending GRAVI it doesn't have. Which wallet it is, if any, is public on-chain (right after the protocol account's fields), and the exemption applies to that wallet's baskets only.

7. Assets

133 assets are listed today:

- **Tokenized stocks and ETFs** issued on Solana (xStocks and others): NVIDIA, Apple, Tesla, the S&P 500, the Nasdaq 100, gold and more.
- **Pre-IPO tokens**: Anthropic, OpenAI, Anduril, Polymarket and others.
- **Crypto**: SOL, bitcoin (wBTC, cbBTC), ETH, HYPE, ZEC, Jupiter, TAO and more.
- **Memecoins**: from BONK and WIF to stonkfun's tokens.

An asset is listed only if it has a working market on Solana. A token whose own rules let its issuer take a cut of every transfer is accepted only up to 4%, and the program checks that cap on every deposit. Tokens with an active transfer hook, which hands every transfer to another program, are refused.

Tokenized stocks are issued by third parties, who can freeze or claw back their tokens under their own terms; GraviFi can't prevent that, only keep redemption of everything else open (section 2.2). Their issuers don't offer them to US persons.

7.1 VERIFIED AND UNVERIFIED

The assets above are **GraviFi Verified**: chosen by the team. A creator can also ask for any other Solana token with a market (open listing). The app checks it first and refuses, whatever else it has going for it, a token that could jam or empty a vault: an active transfer hook, a transfer fee over 4%, accounts that start frozen, a token that can't move or is paused, and freeze, clawback or pause powers held by anyone but a known stock, gold or stablecoin issuer. Leveraged and inverse products are allowed, with a warning that they lose value when the market swings. Beyond that it needs a live price, $10,000 in its pools, 6 hours of trading, and a $150 trade either way within 10% of the market. A token that passes is offered to everyone as **Unverified** and registered on chain in the transaction of the first basket that uses it, signed by the app's listing key (the lister, which the program lets list assets and nothing else); that creator pays the record's rent.

Unverified tokens carry a badge and a one-line note wherever they appear, and so does every basket holding one; Explore and Pulse can be narrowed to baskets of Verified tokens. The rebalancer gives them the wider limits memecoins get, and backs off a basket whose slices keep failing on one. GraviFi doesn't fix or refund problems an Unverified token causes. The app re-checks them in turn every day: one that fails the rules for 3 days running is no longer offered for new baskets (baskets holding it keep it), and one Jupiter hasn't priced for a day is valued at zero so its baskets stay tradable. A token that meets the Verified bar for a week (for stocks from a

known issuer, a $150 trade within 2-3%; for others, also $1M market cap and $100,000 in pools) can be promoted by the team. Every token shows its contract address.

8. Who controls what

There is no governance and no vote. Control is split so that no role can reach holders' funds:

ROLE	CAN	CAN'T
Holders	Redeem at any time, send or trade their shares	
A basket's creator	Change their basket's weights, assets, rebalancing, fee (within the caps), stop losses, split; set how wallets show its shares (name, ticker, picture); set an immutable basket's stop losses before its first deposit	Touch the vault's tokens; once the basket is immutable, anything but how wallets show its shares
The lister (the app's listing key)	List a token a creator picked, inside the creator's own transaction, once it passed the app's checks	Anything else: delist, change a setting, move a token; or list a token whose new accounts start frozen, that can't move, or that's paused
The protocol authority (the team)	List assets, name the lister, set fees and the creation burn within the ceilings, name the one burn-exempt team wallet, match assets to Pyth feeds, name the keepers, set how wallets show any basket's shares, and upgrade the program	Through any of its settings: move a vault's tokens, block a redemption, exceed the fee or creation-burn ceilings, send less than half of the protocol fee to the burn (outside an invited wallet's half fee)
The price service (its own key)	Post the prices rebalances, buybacks and stop losses are checked against, within the bounds above, read from Jupiter by itself	Anything else: trade, move a token, change a setting
The keeper (the app's server)	Run rebalance and buyback slices within the program's checks, each paying it back at most $0.05 for its network costs; ask the price service to post a price (anyone may ask the program to check a stop); co-sign an invited wallet's invite record	Post or choose a price; move a vault's tokens outside a slice, or accept less back than the posted price allows; take more than $0.05 from a slice; give a wallet half fees for more than 31 days at a time
The burn address	Buy GRAVI and burn it	Anything else: it has no key

The program and protocol authority are held today by one team key, and upgrading a program can change what it does. Prices and trades take two more keys, held apart: the price service's key posts prices, the app server's key trades. Taking value through a false price needs both. The price key alone can post a price only within the bounds above, which could send a rebalance the wrong way or fire a stop early, but it can't trade; the trading key alone trades only at the prices the price service posted. Holders trust the team with these keys today. That is why the next steps (section 10) are a multisig, so no single key can upgrade it, and a verified build, so anyone can check that what runs on-chain is the published code.

9. Risks

GraviFi is new software. Before using it, know that:

- **No independent audit yet.** The program has passed an internal security review and its test suite; an independent audit is next. Only put in what you can afford to lose.
- **Key risk.** The team holds the keys that upgrade the program and post the prices rebalances are checked against (section 8). Posted prices are bounded (section 6), but until the multisig lands and prices and trading use separate keys, a lost or misused key could cost basket holders money.
- **Issuer risk.** Tokenized stocks and pre-IPO tokens depend on their issuers, who can pause, freeze or claw back their tokens. A holder can still redeem everything else and leave such a token in the basket (section 2.2).
- **Market risk.** Baskets move with what they hold. Thinly traded tokens can cost more to buy and sell, and the app shows that cost before you sign.
- **Buys take more than one transaction.** The swaps land first, then the mint. If a swap fails, no shares are minted and the tokens already bought stay in your wallet.
- **GRAVI** is a token on an open market, with no rights attached. Its price can go to zero.

Nothing in this paper is investment advice, or an offer to sell GRAVI or any other asset.

10. Roadmap

1. **Independent security audit.**

2. **Multisig** for the program and protocol authority, and a **verified build**, so anyone can check that the program on-chain is exactly this code.
3. **Separate keys** for posting prices and for trading, and Pyth checks for more assets (the tokenized stocks' feeds) as Pyth keeps them updated on chain.
4. **Baskets of up to 16 assets.**

Addresses

Program	`HHmD4ZP5he9GrcmcNrpjahyK1dWh2MhMwmVpiPRJsw3G`
GRAVI	`TRGNtERnbRtVkaMJ1MqU4DzSEkHrtd2qFesh6Rgravi`
Burn address	`FkMEzsnij3aN9njxcjmGxi1DK43WChBtRWorvXgptUJr`
Team GRAVI wallet	`FK9x1PgAqunjtm57DCSpWLuwFmNEArqbF9aieb2VgnL1`
Team GRAVI lock (Streamflow)	`GxtRDjGXxHQTMRh6FX1qte7yvhAP2tTkHKZrhswRhLyp`
Treasury	`CmBPHj38TGpjGLxgAVYvEYETBM4WYqK6GtBgksgKAym2`
Protocol authority	`B95P86aojwMui9aKFdwMjycbML3QFh7jgJ2VftSj9YJ5`

Every address can be checked on Solscan.

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